

CSCE 647 IMAGE SYNTHESIS · MODULE 05 · LECTURE NOTES

Monte Carlo Integration

Estimating an integral with random numbers, and knowing how wrong you are.

MODULE OUTCOMES

- State the Monte Carlo estimator and prove it is unbiased.
- Derive the $1/\sqrt{N}$ convergence rate and explain its consequences.
- Distinguish bias from variance, and consistency from unbiasedness.
- Explain why Monte Carlo beats quadrature in high dimensions.
- Build the validation suite you will use for the rest of the course.

1 The estimator

We want $\int f(x) dx$ over some domain, and we cannot do the integral analytically. Monte Carlo integration replaces it with an average of random evaluations.

$$F_N = (1/N) \sum_{i=1}^N f(X_i) / p(X_i), \quad X_i \sim p$$

Draw N samples from a probability density p of your choosing, evaluate f at each, divide by the density there, and average. **The division is the entire trick:** regions sampled more often contribute proportionally less per sample, which exactly compensates for the uneven sampling.

1.1 Why it is unbiased

$E[F_N] = E[(1/N) \sum_i f(X_i)/p(X_i)]$	
$= (1/N) \sum_i E[f(X_i)/p(X_i)]$	linearity of expectation
$= E[f(X)/p(X)]$	identically distributed
$= \int f(x)/p(x) p(x) dx$	definition of expectation
$= \int f(x) dx = F$	the p cancels

The requirement is only that $p(x) > 0$ wherever $f(x) \neq 0$. If p is zero somewhere f is not, that region is never sampled, its contribution is never counted, and the estimator is biased.

UNBIASED AT $N = 1$, WHICH IS WORTH SITTING WITH

The derivation never assumed N was large. A *single* sample $f(X)/p(X)$ is an unbiased estimate of the entire integral.

That sounds absurd and is correct. One sample has the right expected value and enormous variance: run the single-sample estimator many times and the average of those runs converges to F , even though any individual run may be wildly off.

This separation is the central idea of the whole subject. Correctness and usefulness are different properties. The estimator is correct from the first sample; making it *useful* is entirely a question of reducing variance, which is what Modules 06, 08, and 09 are about.

THE CHOICE OF p AFFECTS VARIANCE ONLY, NEVER CORRECTNESS

The proof places almost no constraint on p . Any density positive where f is non-zero yields an unbiased estimator.

So choosing p badly does not make your renderer *wrong*. It makes it *slow* — sometimes by factors of thousands, which is indistinguishable from wrong in practice, but the distinction matters for debugging: a noisy image and an incorrect image are different problems with different causes.

The ideal p is proportional to f , which would give zero variance. That is unattainable — normalising it requires knowing $\int f$, which is the problem we started with — but it is the target that importance sampling aims at.

Module 06 is entirely about choosing p well. This result is why that effort is worthwhile and why it cannot break correctness.

2 Variance and the convergence rate

$$V[F_N] = V[f(X)/p(X)] / N \quad \Rightarrow \quad \sigma[F_N] \propto 1/\sqrt{N}$$

The variance of the mean of N independent samples is the variance of one sample divided by N . The standard deviation — which is what the eye sees as noise — therefore falls as the square root of the sample count.

To achieve	Multiply samples by	Render time
Half the noise	4	4×
One quarter the noise	16	16×
One tenth the noise	100	100×
One more decimal digit	100	100×
Two more decimal digits	10,000	10,000×

THE RATE IS BRUTAL, AND THE DIMENSION-INDEPENDENCE IS WHY WE ACCEPT IT

The discouraging half: noise falls slowly. Going from 'visibly noisy' to 'clean' in a difficult scene is routinely a factor of a hundred in time, and no amount of engineering changes the exponent — only the constant.

The decisive half: the rate is $1/\sqrt{N}$ *independent of dimension*. Deterministic quadrature does not have this property. A trapezoid rule in d dimensions converges as $N^{-2/d}$: excellent at $d = 1$, useless at $d = 20$, where the exponent is -0.1 .

A path with k bounces is an integral over roughly $2k$ dimensions — two angles per scattering event — and the Neumann series of Module 02 sums over all k , so the effective dimension is unbounded.

Monte Carlo is therefore not a convenient choice here. It is the only method whose convergence does not collapse entirely, and that single fact determines the shape of the entire field: since the exponent cannot be improved, essentially all research effort goes into reducing the *constant* — the variance of a single sample.

3 Vocabulary, used precisely

Term	Definition	Rendering example
Unbiased	$E[\text{estimate}] = \text{true value}$, for every N including $N = 1$.	Path tracing with Russian roulette.
Consistent	The estimate converges to the true value as $N \rightarrow \infty$.	Photon mapping — biased at any finite photon count, converging only as the count grows without bound.
Biased	$E[\text{estimate}] \neq \text{true value}$.	Any fixed bounce limit; any clamp on sample values; irradiance caching.
Variance	Spread of the estimate about its mean.	Visible directly as image noise.

Unbiasedness implies consistency, but not the reverse: a consistent estimator can be wrong at every finite N and still converge eventually. In practice, 'biased but consistent' usually means *systematically wrong in a way that gets smaller as you work harder*, which is often an acceptable trade — provided it is a decision rather than an accident.

	Variance	Bias
Appearance	Noise: speckle, grain, isolated bright pixels.	Systematic error: uniformly too dark, over-blurred, an effect missing entirely.
Per-pixel behaviour	Different in every pixel; changes with the random seed.	Identical every run; independent of the seed.
With more samples	Decreases as $1/\sqrt{N}$.	Does not decrease.
Averaging many renders	Converges to the correct answer.	Converges to the <i>wrong</i> answer.
Aesthetics	Ugly and honest.	Frequently prettier than the truth, which is what makes it dangerous.

The practical diagnostic is the averaging test: render the same scene ten times with different seeds and average the results, then compare against a single very long render. If they agree, you have variance. If they differ, you have bias, and the difference image shows you where.

4 Build the validation suite now

This section is the most practically valuable in the module. Build these five tests today, before writing the integrator, and keep them running for the rest of the course.

THE WHITE FURNACE TEST

Setup: a white Lambertian object, albedo exactly 1.0, inside a uniform environment emitting radiance exactly 1.0 in all directions.

Expected result: every pixel is exactly 1.0, and the object is *invisible* — it reflects precisely as much as it receives, so nothing distinguishes it from the background.

What it catches: a missing or extra cosine factor; a missing π or $1/\pi$; an incorrect PDF; a BRDF that does not conserve energy; a broken direction-sampling routine; an incorrect hemisphere measure. That is most of the ways a renderer can be wrong.

And the failure is diagnostic rather than merely alarming. The ratio to 1.0 usually names the bug: a result of 0.5 points at the cosine; π or $1/\pi$ at a normalisation; a value that varies with surface orientation at the hemisphere measure. Run it at several albedos — the deviation should scale predictably.

Test	What it checks	Cost to build
Hemisphere = 2π	Your solid-angle measure and any uniform direction sampler.	Three lines. There is no excuse for not having it.
White furnace	Energy conservation end to end. See above.	A scene file and an assertion.
χ^2 test on every sampler	That each sampling routine actually draws from the PDF it claims. Histogram a million samples and compare against the analytic density.	An afternoon, and it will find a real bug. A sampler/PDF mismatch is <i>invisible</i> in an image — it produces a plausible, converged, wrong result.
Convergence plot	RMSE against a converged reference, versus sample count, on log-log axes. See below.	A script. Run it on every integrator you write.
Reference comparison	The same scene rendered in PBRT, differenced against yours.	Scene conversion, once. Then it is free forever.

THE SLOPE OF A CONVERGENCE PLOT DIAGNOSES THE PROBLEM

Slope -0.5 : correct behaviour. Pure variance, converging at the theoretical rate.

Slope -0.5 , then flattening to horizontal: bias. The estimator is converging, but not to the reference. Look for a truncated path depth, a clamp, a missing term, or a reference that is itself not converged.

Slope shallower than -0.5 : your samples are not independent. The usual causes are a poor random number generator or the same seed used across pixels or frames.

Slope steeper than -0.5 : you are using a low-discrepancy sequence, which genuinely does better than $1/\sqrt{N}$ on sufficiently smooth integrands. This is legitimate, and Module 06 explains it.

4.1 Fireflies

The first pathology you will meet is a single pixel, or a scattering of them, far brighter than everything around it.

- **Cause:** one sample with an enormous f/p ratio — a direction where the integrand was large but the sampling density happened to be near zero. The estimator divides by that tiny density and produces a huge value.
- **Why they persist:** they are rare and extreme, so they dominate the variance. Averaging many samples removes them only slowly, and a single firefly can survive thousands of samples.
- **Clamping** — capping sample values at some maximum — removes them instantly and **introduces bias**, because the energy discarded was real. Most production renderers clamp. **The problem is not that they clamp; it is that it is often an accident rather than a decision.** If you clamp, say so, and report the threshold.
- **The unbiased responses** are better importance sampling (Module 06), multiple importance sampling (Module 08), and firefly-aware denoising (Module 12). Each attacks the cause rather than the symptom.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapter 2, Monte Carlo Integration	The estimator, variance, and the sampling machinery, with proofs. The primary reading.
Veach thesis — Chapter 2, Monte Carlo Integration	More careful than most treatments on what unbiasedness does and does not guarantee.
TU Wien Rendering — Monte Carlo lectures	The intuition for why dividing by the PDF works, built up slowly. Worth watching before reading PBRT.
PBRT — the χ^2 test in the test suite	A working implementation of the sampler validation in §4. Copy the approach.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Estimate π by Monte Carlo. Plot error against N on log-log axes and confirm the slope is -0.5 . Keep the plotting script — you will reuse it all semester.
- 2 Integrate a function you know analytically, using three different densities p. Confirm all three converge to the same value and report their variances. This makes §1's claim concrete.

- 3 Deliberately choose a p that is zero where f is non-zero. Show the estimator converges confidently to the wrong answer.
- 4 Implement the hemisphere test and the white furnace test. Then introduce each of these bugs in turn and record what the furnace test reports: drop the cosine; drop the $1/\pi$; use an albedo of 1.1; use the wrong PDF for cosine sampling.
- 5 Write a χ^2 test for a sampling routine. Then deliberately mismatch the sampler and its stated PDF by a constant factor and confirm the test catches it while the rendered image does not look obviously wrong.
- 6 Produce a convergence plot with an artificial bias introduced — clamp all samples at 5.0 — and confirm the curve flattens.
- 7 Create a scene that produces fireflies. Measure the variance with and without clamping, and measure the energy lost to the clamp. Report both numbers.
- 8 Compare the variance of estimating a 10-dimensional integral by Monte Carlo against a tensor-product trapezoid rule with the same number of evaluations. Report both errors.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the Monte Carlo estimator and explain what the division by p is doing.
- 2 Prove unbiasedness. What is the only condition on p ?
- 3 Why is a single-sample estimate unbiased, and what does that tell you about the relationship between correctness and usefulness?
- 4 How does the choice of p affect the result, and what would the ideal p be?
- 5 State the convergence rate. How many more samples for half the noise, and for one more digit?
- 6 Why is dimension-independence decisive for rendering specifically?
- 7 Distinguish unbiased, consistent, and biased, and give a rendering example of each.
- 8 How do you tell bias from variance by looking at images?
- 9 What is the white furnace test, and name four bugs it catches.
- 10 What does a convergence plot that flattens out tell you?